

Notes: Cuciniello, Michelacci, and Paciello (JPE, 2025)

Subsidizing Business Entry in Competitive Credit Markets

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1 Big picture

- **Question.** How should governments subsidize business entry when start-ups are financially constrained and borrow in competitive credit markets?
- **Core distinction.** The paper contrasts **ex ante** subsidies paid before business formation with **ex post** reimbursements of start-up expenditures.
- **Main mechanism.** Ex ante subsidies reduce the amount start-ups must borrow, lowering leverage and bankruptcies. Ex post reimbursements arrive after debt has already been issued, so leverage ratchet effects prevent firms from using the full transfer to reduce debt.
- **Debt overhang logic.** A firm that repays debt raises the value of outstanding debt, partly transferring resources to creditors. Equity holders therefore underuse cash flows and reimbursements for deleveraging.
- **Competition effect.** Entry subsidies increase business creation and competition. This lowers profitability and can raise bankruptcy among incumbent firms.
- **Quantitative setting.** The model is calibrated to Northern and Southern Italy, where the South has lower productivity, higher entry, higher exit, higher leverage, and higher bankruptcy rates.
- **Optimal policy.** The optimal subsidy to business creation in the South is entirely paid ex ante and raises welfare by about 2% of consumption.
- **Policy failure.** Paying the same subsidy ex post can reduce welfare because it raises bankruptcies and crowds out equity rather than debt.
- **Empirical check.** The paper uses evidence from Italy's *I Stay in the South* subsidy and firm reimbursements to validate the reimbursement and leverage-ratchet channel.
- **Takeaway.** Subsidy timing is not an implementation detail. In competitive credit markets with long-term debt, paying the subsidy before entry can be welfare improving, while reimbursing expenditures after entry can be welfare reducing.

2 Data and measurement

2.1 Italian regional data

- The calibration targets differences between Northern and Southern Italy in firm dynamics and regional characteristics.
- The South has lower GDP per capita, higher business exit, higher bankruptcy rates, higher business entry, higher leverage, and larger idiosyncratic shocks.
- The risk-adjusted cost of credit is similar in North and South, supporting the model assumption of an integrated competitive credit market.

Interpretation: The North–South comparison is useful because the South has exactly the conditions in which entry subsidies are often used: low productivity, high leverage, high exit, and weak local business dynamics.

2.2 Firm entry, exit, and bankruptcy

- Business entry is the fraction of start-ups relative to total employment.
- Business exit is the fraction of firms that exit during a year.
- Bankruptcy is measured by exits with bad loans or formal bankruptcy procedures.

- Firm leverage is financial debt over total value added.

Interpretation: The model distinguishes ordinary exit from inefficient bankruptcy. A central welfare cost arises when firms with positive continuation value are inefficiently liquidated because of debt and limited liability.

2.3 Government reimbursements and the ratchet channel

- The paper uses Italian firm data on reimbursements to study how firms use government transfers paid after expenditure.
- It estimates whether reimbursements reduce bank debt and increase employment.
- The key empirical question is whether one euro of reimbursement reduces debt by one euro.

Interpretation: If firms face a leverage-ratchet problem, ex post reimbursements should reduce debt by less than one-for-one and may instead increase payouts or equity value.

2.4 I Stay in the South subsidy

- The paper studies the Italian *I Stay in the South* (ISS) program, which subsidizes start-ups in the South.
- Eligible firms receive a mix of grants and subsidized loans.
- The empirical design compares firms or provinces exposed to ISS with less-exposed counterparts.

Interpretation: ISS provides an empirical benchmark for the model's policy experiment and tests whether the simulated policy can reproduce observed effects on creation, exit, leverage, productivity, and firm size.

3 Models and empirical specifications

3.1 Firm value and debt

Firms finance entry with long-term non-state-contingent debt. Let b denote the debt-value ratio and μ market profitability. Equity value can be written as

$$v(b, \mu), \tag{1}$$

with bankruptcy occurring when the value of continuing is too low relative to debt obligations.

Interpretation: The key state variable is debt relative to value. Higher debt raises bankruptcy risk and reduces incentives to invest or repay debt.

3.2 Financial constraint and leverage ratchet

Let $\lambda(b, \mu)$ be the multiplier on the financing constraint. When the firm is financially unconstrained,

$$\lambda(b, \mu) = 0. \tag{2}$$

When the constraint binds,

$$\lambda(b, \mu) > 0. \tag{3}$$

Interpretation: Even financially unconstrained firms do not voluntarily buy back debt because the gains from lower bankruptcy risk accrue partly to creditors. This is the leverage-ratchet effect.

3.3 Ex ante subsidy

An ex ante subsidy lowers the amount that the entrepreneur needs to borrow at entry:

$$b_0^{ex\ ante} < b_0^{no\ subsidy}. \quad (4)$$

Interpretation: Ex ante transfers directly reduce start-up leverage and therefore reduce bankruptcy risk.

3.4 Ex post reimbursement

Let R be a reimbursement after entry and $\Delta(b)$ the amount of debt repaid. A one-for-one debt reduction would have

$$\Delta(b) = R. \quad (5)$$

But with leverage ratchet effects,

$$0 \leq \Delta(b) < R. \quad (6)$$

Interpretation: The firm does not use the full reimbursement to repay debt because debt repayment partly transfers value to existing creditors.

3.5 Empirical ratchet regression

The reimbursement regression is summarized by

$$Y_{j,t} = \alpha_j + \gamma_t + \beta_R R_{j,t} + \varepsilon_{j,t}, \quad (7)$$

where Y is bank debt or employment and $R_{j,t}$ is the reimbursed amount.

Interpretation: A negative coefficient for bank debt that is smaller than one in absolute value indicates that reimbursements do not fully reduce debt.

3.6 Welfare objective

Welfare is the present discounted value of aggregate consumption net of living costs:

$$W_0(T) = \int_0^T e^{-rt} \left[\left(\frac{Y_t - Y}{C} \right) - \left(\frac{I_t - I}{C} \right) - \left(\frac{\mathcal{E}_t - \mathcal{E}}{C} \right) - \left(\frac{H_t - H}{C} \right) \right] dt. \quad (8)$$

Interpretation: Welfare gains come from higher output and lower inefficient bankruptcy costs, but subsidies also affect investment, entrepreneurial leisure costs, living costs, and spatial reallocation.

3.7 Difference-in-differences ISS regression

The empirical ISS design is summarized by

$$Y_{i,t,a} = \alpha_i + \gamma_t + \delta_a + \beta_1 \text{EligibleToSubsidy}_{i,t,a} + \beta_2 \text{SouthIncumbent}_{i,t,a} + \varepsilon_{i,t,a}. \quad (9)$$

Interpretation: The coefficients compare eligible start-ups and South incumbents before and after ISS, controlling for province, year, age, and interaction effects.

4 Identification and empirical design

4.1 What identifies the mechanism

- The theoretical mechanism compares the same subsidy amount paid at different dates.
- Ex ante subsidies affect the initial financing need before debt is issued.
- Ex post reimbursements arrive after debt is already outstanding and are distorted by the leverage ratchet.
- Empirical reimbursement regressions measure whether actual reimbursements reduce debt less than one-for-one.

Interpretation: The paper’s key identification insight is that timing changes the debt-equity trade-off even when the subsidy amount is unchanged.

4.2 Why general equilibrium matters

- Entry subsidies increase competition and reduce profitability.
- Lower profitability raises incumbent bankruptcy risk.
- Workers move across provinces in the long run, changing local labor supply, wages, and living costs.
- The optimal subsidy trades off start-up deleveraging benefits against incumbent shakeout and excessive entry.

Interpretation: A partial-equilibrium evaluation would overstate the benefits of subsidies if it ignored incumbent bankruptcies and spatial reallocation.

4.3 What the paper cannot fully prove

- The model abstracts from some details of financial contracts and bank bargaining.
- The calibration relies on matching regional moments and policy effects rather than a randomized national experiment.
- Nondistortionary taxation is assumed in the optimal subsidy analysis.
- The empirical ISS evidence is used to validate model mechanisms rather than to fully identify welfare.

Interpretation: The paper is strongest as a structural policy analysis disciplined by regional and firm-level evidence.

5 Tables and figures

5.1 Figures 1 and 2: free entry and subsidy timing

Read:

- Figure 1 shows the free-entry equilibrium in the space of profitability μ and initial debt b_0 .
- Figure 2 shows the difference between ex post and ex ante subsidies.
- Ex post subsidies shift the free-entry condition down but leave financing needs unchanged at entry, raising leverage and bankruptcy risk.
- Ex ante subsidies reduce initial financing needs and can lower leverage and bankruptcy risk.

Economic message: The timing of the subsidy changes whether it reduces debt or mainly reduces equity needs after debt has already been issued.

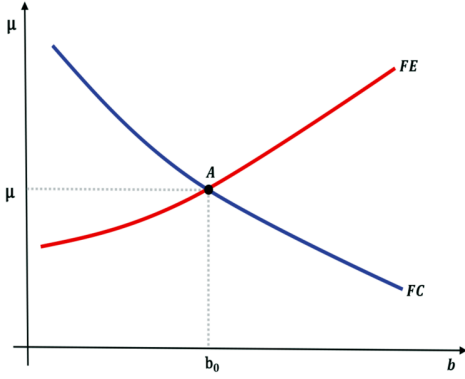


FIG. 1.—Free entry equilibrium: constant μ in each province.

Figure 1: free-entry equilibrium

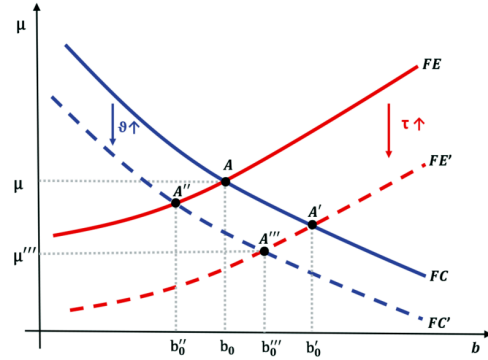


Figure 2: ex post versus ex ante subsidy

5.2 Figures 3 and 4: value function and policies

Read:

- Figure 3 shows equity value and the Lagrange multiplier as functions of the debt-value ratio.
- Figure 4 reports dividend payouts, new debt issuance, investment, and the price of debt.
- The financial constraint binds over an intermediate range of leverage.
- Investment and payouts respond sharply to leverage and bankruptcy thresholds.

Economic message: The policy functions illustrate the leverage-ratchet mechanism: high leverage distorts payout, debt issuance, investment, and debt pricing.

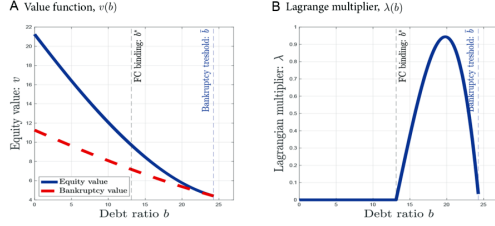


FIG. 3.—Value function and Lagrange multiplier. The parameter values are as in table 2 for the Southern economy.

Figure 3: value and multiplier

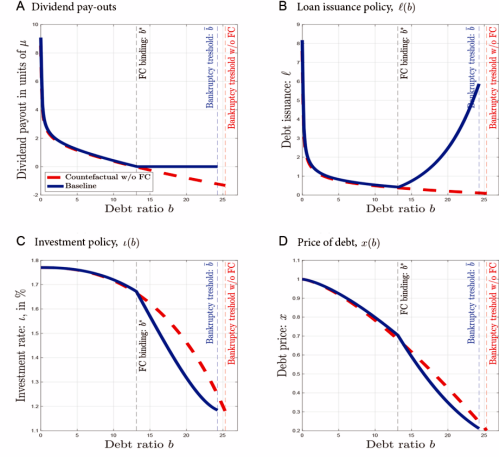


Figure 4: policy functions

5.3 Figures 5 and 6: ex post reimbursement and debt payback

Read:

- Figure 5 shows that the fraction of the subsidy used to reduce debt depends on the initial debt-value ratio.
- Figure 6 shows that larger reimbursements are not necessarily used more intensively for debt payback.
- For highly leveraged firms, only part of the reimbursement is used to repay debt.
- The optimal debt payback remains below the reduction that would fully undo the financial constraint.

Economic message: Ex post transfers do not translate mechanically into deleveraging. They can crowd out firm equity and leave bankruptcy risk elevated.

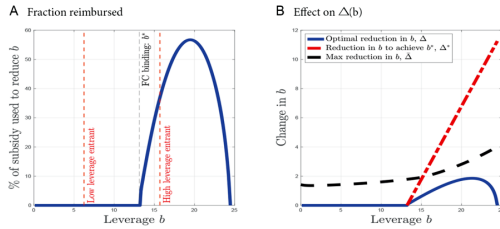


FIG. 5.—Debt payback and initial debt-value ratio b . The parameter values are as in table 2 for the Southern economy. The ex post subsidy is equal to €90,000, the average size of

Figure 5: debt payback and initial leverage

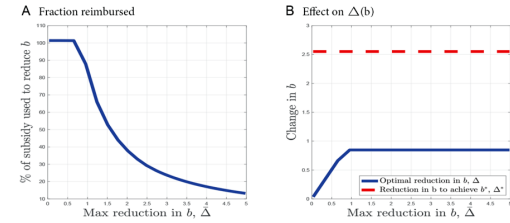


FIG. 6.—Debt payback and size of subsidy $\bar{\Delta}$. The parameter values are as in table 2 for the Southern economy. The firm has a debt-value ratio corresponding to the rightmost vertical dashed line in figure 5A (high leverage entrant).

Figure 6: debt payback and subsidy size

5.4 Tables 1 and 2 plus Figure 7: calibration targets

Read:

- Table 1 documents North–South differences in productivity, exit, bankruptcy, entry, leverage, debt cost, shocks, and firm size growth.
- Table 2 reports calibrated parameters and targeted moments.
- Figure 7 compares model and data age profiles for leverage, exit, and bankruptcy rates.
- The model reproduces the key differences in business failure and leverage between North and South.

Economic message: The calibration is designed to match both levels and age profiles of firm dynamics, which are central for evaluating entry subsidies.

	Population Share (1)	GDP per Capita (2)	Business Exit (%) (3)	Bankruptcy Rate (%) (4)	Business Entry (%) (5)	N Firms per Capita (6)	Leverage Ratio (7)	Debt Cost r_e (%) (8)	SD of Idiosyn- cratic Shocks, σ (9)	Relative Employment Size, 11–14 Years (10)
Italy	1.00	1.00	8.06	5.32	.57	.06	1.78	1.08	.26	1.17
North	.65	1.19	7.75	5.18	.46	.05	1.77	1.10	.24	1.16
South	.35	.66	8.88	5.71	1.08	.09	1.81	1.07	.31	1.18

NOTE.—The table shows averages in Italy, North, and South calculated over the period 2007–15, with the exception of the population share and GDP per capita, which are averaged over the period 2009–15. The South encompasses the eight Italian regions traditionally known as Mezzogiorno, and the North encompasses all the other regions. Italian GDP per capita is normalized to 1. Business exit is the fraction of companies exiting during a year. The bankruptcy rate is the fraction of firms in a year that exit leaving behind some bad loans or after a formal bankruptcy procedure. Business entry is the ratio of start-ups to total employment. Number of firms per capita is the number of companies over total employment. The leverage ratio is the ratio of total financial debt of companies in the area over their total value added. The debt cost r_e is the average risk-adjusted real cost of credit discussed in the text. The SD of idiosyncratic shocks, σ , is the average standard deviation of firm-level idiosyncratic shocks in the region. The relative employment size is the average employment size of a 14-year-old firm in the region divided by the size of an 11-year-old firm. Appendix G describes the data in more detail.

TABLE 2
PARAMETER VALUES

	(1)	(2)
A. Parameter Values		
Population size (ζ_N, ζ_S)	.65, .35	
Firm discount rate (r)	.030	
Financiers' discount rate (r_e)	.011	
Elasticity of substitution (ν)	4.5	
Minimum technology at entry (z_e^N, z_e^S)	−.179, −.50	
Maximum technology at entry (z_e^N, z_e^S)	.038, .42	
Probability of z^a at entry (q_N, q_S)	.81, .43	
Idiosyncratic risk (σ_N, σ_S)	.25, .31	
Entry investment cost (k_N, k_S)	33.5, 14.4	
Entrepreneur leisure cost (χ_N, χ_S)	.03, .03	
Initial entrepreneur's wealth (ϖ_N, ϖ_S)	13.7, 9.4	
Entrepreneur death rate (δ_N, δ_S)	.026, .029	
Debt coupon (κ)	$r + \delta$	
Debt maturity arrival rate (ρ)	1/30	
Recovery rate with renegotiation (α_N, α_S)	.65, .63	
Renegotiation probability (ϕ_N, ϕ_S)	.46, .53	
Scale of congestions ($\bar{h}_N, \bar{h}_S$)	.47, .07	
Congestion elasticity to workforce (η)	3.0	
Curvature adjustment cost function (θ)	2	
Adjustment cost level (ζ_N, ζ_S)	.010, .004	
Worker mobility rate (ψ)	.069	
B. Targeted Moments		
	Data Value	Model Value
Population share, North and South	.65, .35	.65, .35
Yearly wealth return	.030	.030
Risk-adjusted interest rate on debt	.011	.011
Average value in INVIND	4.5	4.5
Average labor productivity (y_i/ℓ_i), North and South	1.19, .66	1.19, .66
Average exit rate, age 0, North and South	.109, .098	.104, .098
Average exit rate, age 1, North and South	.117, .104	.115, .104
SD of shocks, North and South	.25, .31	.25, .31
Average firm size (ℓ_i/m_i), North and South	20.00, 11.11	20.00, 11.11
Leverage ratio, age 12–14 years, North and South	1.74, 1.73	1.76, 1.74
Leverage ratio at entry, North and South	2.74, 2.05	2.74, 2.05
Overall average exit rate, North and South	.026, .029	.025, .029
Debt cost to entrepreneur absent default (φ)	1.00	1.00
Leverage ratio at age 4 relative to age 0	.78, .88	.76, .89
Bankruptcy recovery rate, North and South	.200, .200	.207, .208
Bankruptcy rate, age 12–14 years, North and South	.042, .051	.044, .055
North workforce and living costs over GDP	1.00, .33	1.00, .33
Long-run elasticity of workforce to wages	2.6	2.6
Investment elasticity to marginal capital value	2	2
Relative employment size, age 11–14 years, North and South	1.14, 1.17	1.15, 1.20
Half-life duration of aggregate shocks (years)	10	10

Table 2: parameter values and targets

5.5 Table 3: reimbursement and the leverage-ratchet effect

Read:

- The amount reimbursed reduces bank debt, but by much less than one-for-one.
- The coefficient on amount reimbursed is about -0.14 , meaning firms use roughly 14% of the reimbursement to repay bank debt.
- Reimbursements also increase employment, consistent with part of the transfer being used for investment

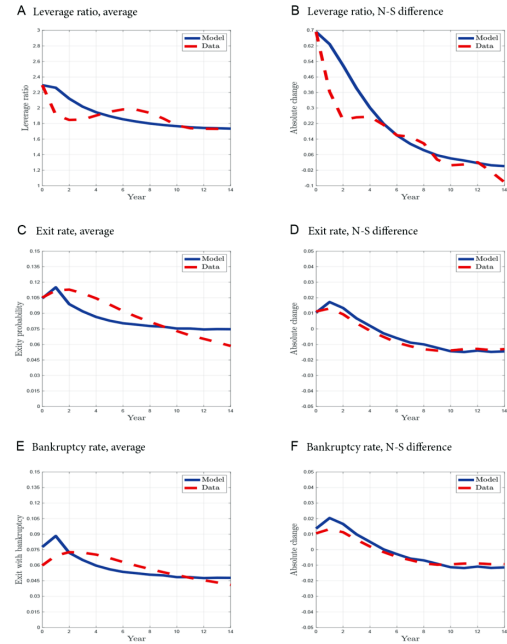


FIG. 7.—Age profiles of business failure and leverage: model versus data. The dashed line corresponds to data for all Italian companies averaged over the 2007–15 period. Leverage is total firm debt over firm value added. The average leverage ratio is weighted across firms using firm value added. The bankruptcy rate is the exit rate of companies that leave behind some bad loans or exit after a bankruptcy procedure. A, C, and E plot cross-region averages obtained by averaging the age profiles specific to each region. $\ell = 0$ – N by their

Figure 7: age profiles and calibration fit

or operations.

Economic message: The empirical reimbursement evidence supports the model’s ratchet effect: ex post subsidies mostly do not reduce debt.

Variable	Bank Debt		Employment	
	(1)	(2)	(3)	(4)
Amount reimbursed (€10,000)	−.14** (.07)		.06* (.03)	
Month of reimbursement		−.13* (.07)		.01 (.04)
Months after reimbursement:				
1 month		−.14* (.07)		.03 (.04)
2 months		−.13* (.07)		.07* (.04)
More than 2 months		−.14** (.07)		.06* (.03)
Observations	28,701	28,701	28,701	28,701
R^2	.75	.75	.55	.55
Firm dummy	Yes	Yes	Yes	Yes
Year-month dummy	Yes	Yes	Yes	Yes
Reimbursement dummy	Yes	Yes	Yes	Yes

NOTE.—The table shows estimates from running the regression (65). The dependent variable is firm bank debt B_{jt} (cols. 1, 2) and firm employment N_{jt} (cols. 3, 4). Data are at a monthly frequency over the period 2017–23. Standard errors (in parentheses) are clustered by reimbursement event.

* $p < .10$.

5.6 Figures 8–11 and Table 4: optimal subsidy and welfare

Read:

- Figure 8 shows the long-run effects of subsidy size on profitability, entry debt, and labor force.
- Figure 9 reports impulse responses at the optimal ex ante subsidy.
- Figure 10 shows that welfare is maximized by a positive subsidy paid entirely ex ante.
- Figure 11 shows that delaying the subsidy increases entry leverage and raises exit rates.
- Table 4 decomposes welfare gains into output, investment, congestion, leisure, transfers, and reallocation components.

Economic message: A well-designed ex ante subsidy improves welfare, while delayed payment reduces or reverses welfare gains by raising leverage and exit.

greater business creation investment in the South (fig. 9C), directly caused

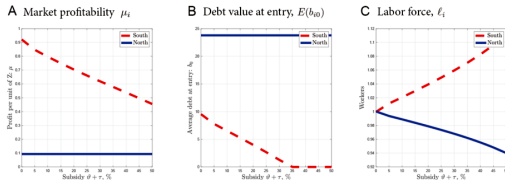


Figure 8: long-run subsidy effects

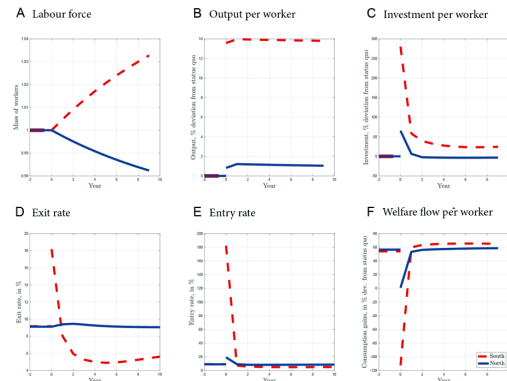


Figure 9: impulse responses

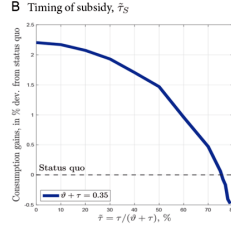
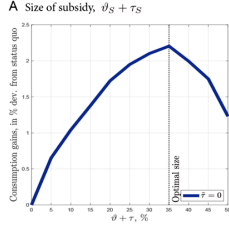


FIG. 10.—Welfare effects of subsidy.

2 percentage points. The optimal subsidy $(\tau_s + \theta_s)k$ amounts to roughly

Figure 10: welfare and subsidy timing

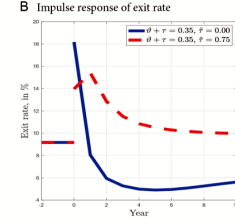
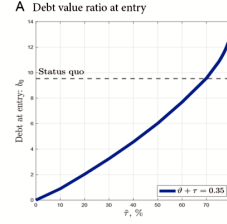


FIG. 11.—Leverage, exit, and timing in South.

Figure 11: leverage, exit, and timing

	Aggregate, W_0 (1)	South, W_0^S (2)	North, W_0^N (3)	Reallocation, $W_0 - (\varsigma_S W_0^S + \varsigma_N W_0^N)$ (4)
A. Optimal Timing, $\tilde{\tau} = 0$				
1. Total welfare	2.20	4.34	.96	−3.10
2. Steady-state welfare only	5.55	8.62	3.81	−6.88
3. Output	4.12	11.86	1.36	−9.10
4. Investment	−5.01	−13.41	−.44	8.84
5. Congestions	3.18	−1.25	4.43	−.00
6. Entrepreneur leisure	−.09	−.45	−.07	.44
7. Net tax transfer	−.00	7.60	−4.33	−3.27
B. Suboptimal Timing, $\tilde{\tau} = .75$				
1. Total welfare	.06	1.11	−.53	−.53
2. Steady-state welfare only	2.54	4.29	1.59	−3.34
3. Output	2.94	8.60	.98	−6.63
4. Investment	−5.18	−14.06	−.32	9.20
5. Congestions	2.35	−.89	3.24	.00
6. Entrepreneur leisure	−.05	−.31	−.05	.30
7. Net tax transfer	.00	7.77	−4.37	−3.40

5.7 Figures 12 and 13: welfare decomposition and Dixit–Stiglitz comparison

Read:

- Figure 12 decomposes welfare gains across counterfactual economies.
- The full model’s welfare gain depends on financial constraints, spatial reallocation, congestion, and aggregate demand.
- Figure 13 compares the baseline economy with a Dixit–Stiglitz benchmark.
- The optimal subsidy differs across economies because the benefits and costs of entry differ.

Economic message: The welfare result is not only a standard product-variety effect. It depends on the interaction between finance, bankruptcy, and spatial equilibrium.

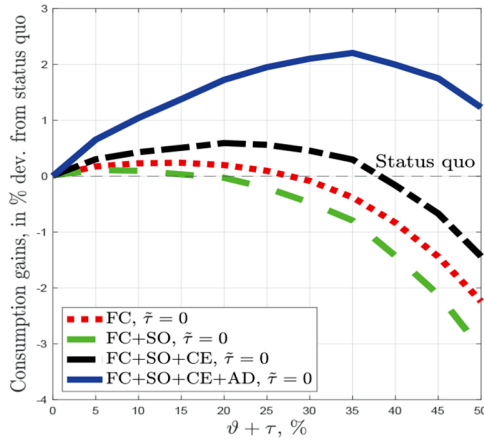


FIG. 12.—Welfare gains decomposition at optimal subsidy. FC refers to an economy where the labor forces ℓ_i 's and the demand shifter Y are at their initial steady-state value and, on impact, the distribution of incumbent firms is unaffected by the change in μ_c .

Figure 12: welfare decomposition

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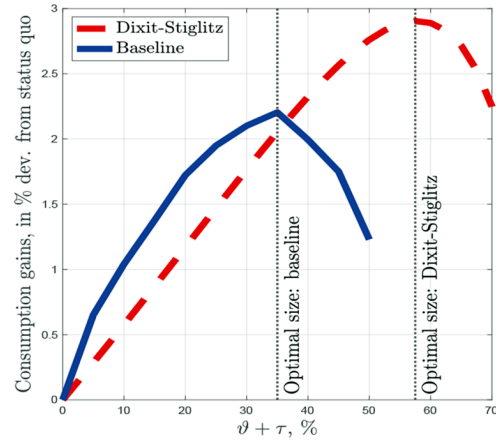


Figure 13: Dixit-Stiglitz comparison

5.8 Tables 5–6 and Figures 14–15: I Stay in the South

Read:

- Table 5 reports difference-in-differences estimates for the ISS policy.
- Eligible start-ups experience higher creation, higher exit, higher leverage, higher productivity, and smaller size.
- Figure 14 shows welfare effects of the ISS subsidy under different incidence assumptions.
- Figure 15 shows that as the incidence of the subsidy changes, subsidized and unsubsidized firms differ in initial debt and exit rates.
- Table 6 shows that model-simulated data reproduce the main signs and magnitudes of the empirical ISS effects.

Economic message: ISS validates the model's prediction that subsidy design can raise creation while also increasing leverage and exit when the policy operates like delayed support.

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TABLE 5
DIFFERENCE-IN-DIFFERENCES EFFECTS OF ISS SUBSIDY

Variable	Creation (1)	Exit Rate (2)	Leverage (3)	Productivity (4)	Size (5)
<i>Eligible-to-Subsidy</i>	.10*** (.01)	.54*** (.21)	.14*** (.04)	.04*** (.02)	-.05*** (.02)
<i>South-Incumbent</i>		.22 (.19)	.08** (.04)	.02* (.01)	-.04** (.02)
Observations	1,045	6,175	6,175	6,175	6,175
R^2	.99	.40	.36	.60	.67
Province dummy	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes
Age dummy	No	Yes	Yes	Yes	Yes
Province $\times$ age dummy	No	Yes	Yes	Yes	Yes
Year $\times$ age dummy	No	Yes	Yes	Yes	Yes

NOTE.—The table shows estimates from running the regression in (72) on the population of companies in which at least one of the founding shareholders is younger than 55 years of age over the period 2010–20. Business creation, productivity, and employment size are the log of firm averages. The business exit rate is in percentage terms (i.e., multiplied by 100). The leverage ratio is in level. Standard errors are in parentheses.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

Table 5: empirical ISS effects

TABLE 6
DIFFERENCE-IN-DIFFERENCES EFFECTS OF ISS SUBSIDY ON MODEL-SIMULATED DATA

Variable	Creation (1)	Exit Rate (2)	Leverage (3)	Productivity (4)	Size (5)
<i>Eligible-to-Subsidy</i>	.17*** (.01)	.45*** (.01)	.04*** (.00)	.04*** (.00)	-.07*** (.00)
<i>South-Incumbent</i>		.32*** (.01)	.03*** (.00)	.04*** (.00)	-.06*** (.00)
Observations	1,045	6,175	6,175	6,175	6,175
R^2	.62	1.00	1.00	1.00	1.00
Province dummy	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes
Age dummy	No	Yes	Yes	Yes	Yes
Province $\times$ age dummy	No	Yes	Yes	Yes	Yes
Year $\times$ age dummy	No	Yes	Yes	Yes	Yes

NOTE.—The table shows estimates from running the regression in (72) on model-simulated data. Business creation, productivity, and employment size are the log of firm averages. The business exit rate is in percentage terms (i.e., multiplied by 100). The leverage ratio is in level. Standard errors are in parentheses.

*** $p < .01$.

Table 6: model-simulated ISS effects

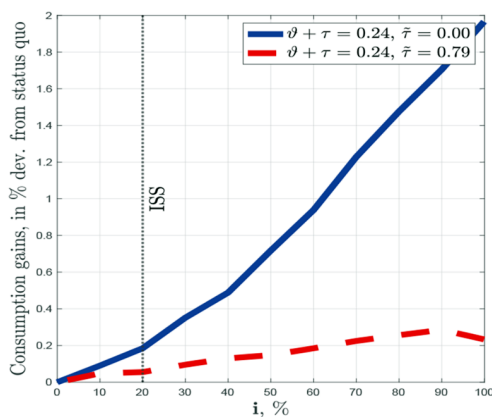


FIG. 14.—Welfare effects of ISS subsidy
Figure 14: ISS welfare effects

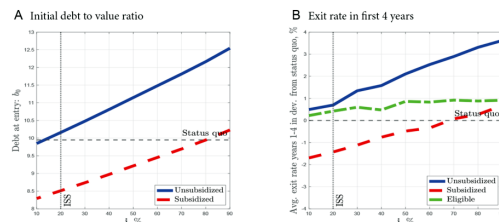


FIG. 15.—Effects on debt and exit of changing incidence of ISS. The size of the subsidy is $\vartheta_s + \tau_s = 0.24$, and the timing is $\bar{\tau} = 0.79$. The incidence of ISS corresponds to the vertical dotted line.

Figure 15: ISS incidence, debt, and exit

6 Short summary

- The paper studies business creation subsidies in a model with financially constrained entrants and competitive long-term debt.
- Firms cannot commit to future funding decisions, creating a leverage-ratchet effect.
- Ex ante subsidies reduce start-up debt directly and reduce bankruptcy risk.
- Ex post reimbursements arrive after debt is issued and are only partly used to repay debt.
- Entry subsidies also increase competition and can raise incumbent bankruptcies.
- The model is calibrated to North–South differences in Italy.
- The optimal subsidy for the South is paid entirely ex ante and raises welfare by about 2% of consumption.
- A delayed or ex post subsidy can create welfare losses.
- Empirical reimbursement regressions show that firms use only a small fraction of reimbursements to reduce bank debt.
- ISS evidence is consistent with the model: subsidies raise creation, leverage, exit, and productivity while reducing firm size.

7 Conclusion

7.1 Core conclusion

Cuciniello, Michelacci, and Paciello show that the design of business entry subsidies matters as much as the subsidy size. In competitive credit markets with long-term debt, subsidies paid before entry reduce start-up leverage and bankruptcy risk, while reimbursements paid after entry can crowd out equity rather than debt and increase bankruptcies.

7.2 Economic intuition

The key intuition is the leverage-ratchet effect. Once debt is outstanding, repaying it increases the value of existing creditors' claims, so equity holders do not internalize the full social benefit of deleveraging. Ex ante subsidies avoid this distortion because they reduce the need for debt before it is issued. Ex post subsidies do not.

7.3 Takeaway

Entry subsidies can be powerful development policy, but their timing is crucial. Policies that look equivalent in a frictionless model are very different once firms finance entry with long-term competitive debt. The paper's lesson is that governments should subsidize entry before firms borrow, not reimburse firms after leverage has already been created.